

1. Overview

The **Data Housekeeping Function** is responsible for purging and cleaning obsolete, irrelevant, or outdated data from the database to ensure optimal performance, security, and compliance with organizational data retention policies.

This function automates the process of **archiving, deleting, and maintaining** data across multiple modules, thereby improving overall system performance, reliability, and scalability.

2. Key Functionalities

2.1 Database Backup

- Perform a **full backup** of the database before executing any archival or purging operations for the older audit logs.
- Ensure **data safety and recoverability** in case of failure during the process.
- Maintain **backup logs** with timestamps and backup file paths for auditing.

2.2 Archive Old Data

- Archive data that exceeds the **defined retention threshold** (e.g., older than 1 year) into designated **archive tables** or databases.
- Only **relevant and validated records** are archived to ensure data consistency and integrity.
- Archive operations are executed in **batches** to minimize performance impact.

2.3 Delete Permanently Deleted Service References

- Identify and **purge records** related to permanently deleted service references to reduce database clutter.
- Maintain referential integrity by ensuring dependent records are also removed.

- Update housekeeping logs to reflect details of purged records.

2.4 Delete Disabled Service and User References as per retention period

- Remove **User, Service, and LOB references** that have been disabled for more than one year.
- This includes associated metadata and logs to ensure minimal residual data.
- Improve database efficiency by cleaning up inactive entities.

2.5 Clean Up Error and Information Logs

- Purge **error, audit, and informational logs** older than **one month** to prevent log table bloat.
- Retain only the most recent and relevant logs necessary for monitoring and troubleshooting.
- Optimize logging tables for post-cleanup to maintain efficient read/write performance.

2.6 Batch Processing

- Implement **batch-wise archival and deletion** to avoid excessive locking and minimize impact on active transactions.
- Typical batch size: **1,0000 rows per iteration** (configurable based on environment).
- Introduce **short delays** (e.g., `WAITFOR DELAY` in SQL or `Task.Delay` in .NET) between batches for smoother performance.

2.7 Table Maintenance and Optimization

- Periodically **purge irrelevant or outdated records** from large operational tables.
- Perform **index reorganization or rebuild** after major deletions to maintain query performance.

- Execute **statistics update** to ensure optimal query execution plans.

2.8 Logging and Auditing

- Log every archival, deletion, and cleanup operation into a centralized **archive_log** table.
- Each log entry should include:
 - Operation type (Archive / Delete)
 - Table name
 - Row count affected
 - Timestamps
 - Execution status (Success / Failure)
 - Error details (if any)
- These logs ensure **traceability and transparency** for audit and compliance purposes.

3. Process Flow

1. **Start Process:** Trigger initiated manually or via scheduled background job.
2. **Backup Phase:** Perform full database backup.
3. **Archival Phase:** Move old data to archive tables in batches.
4. **Purge Phase:** Delete obsolete, disabled, or permanently removed records.
5. **Log Cleanup Phase:** Remove older log entries.
6. **Optimization Phase:** Reorganize and update tables.
7. **Logging Phase:** Record process summary in **archive_log**.
8. **Completion:** Send notification (optional) with process summary.

4. Benefits

Improved Performance

- Removing outdated and irrelevant data enhances query response times and overall database performance.

Storage Optimization

- Significantly reduces unnecessary storage usage by archiving and deleting obsolete data.

Regulatory Compliance

- Ensures alignment with organizational and legal **data retention policies**.

Automation

- Fully automated and **scheduler-driven** process requiring minimal manual intervention.

Data Safety

- Pre-emptive backups and detailed error handling ensure **data integrity and recoverability** in case of failures.

Transparency

- Comprehensive logging provides **end-to-end visibility** of the housekeeping operations.

5. Security and Compliance Considerations

- All data archival and purge operations must follow **access control policies** to prevent unauthorized deletions.
- Ensure **encrypted storage** of archived data if required by compliance standards.

- Validate data consistency between main and archive tables for post-archival.

6. Conclusion

The **Data Housekeeping Function** plays a critical role in maintaining a **clean, optimized, and compliant** database environment.

By automating the archiving, purging, and maintenance tasks, it enhances **performance, reliability, and compliance** while ensuring **data integrity and operational continuity**.

This function is designed to be **scalable, secure, and auditable**, supporting long-term system sustainability and performance excellence.